Name: Himanshi Kaushal
UID: 23BCS12785
Section: KRG-2A

Experiment 7

* Approach:

We need the cheapest flight from src to dst with atmost k stops.
A normal shortest path algorithm cannot be used directly because here cost is important, but the number of stops is also restricted. So we use Bellman-Ford approach.

Use Bellman Ford for only k+1 relaxations. Because if there are atmost k stops, then the route can use atmost k+1 edges, so we relax all flights exactly k+1 times.

* Pseudocode:

```
function cheapestFlight(n, flights, src, dst, k):
        dist [0...n-1] = INF
        dist[src] = 0
        for i=0 to k:
            temp = copy of dist
            for each (u, v, w) in flights:
                if dist [u] != INF and dist [u]+
                w < temp [v]:
                    temp [v]= dist [u] + w
```

```
dist = temp
if dist[dst] == INF:
    return -1;
else:
    return dist[dst]
```

* Time Complexity:

$O(k \times m)$

where k = no of stops

m = no. of flights

Space Complexity:

$O(n)$

where n = no. of cities.